BUSINESS INTELLIGENCE LAB

EXPERIMENT 7

Integration of Advanced Analytics with BI Reports using Microsoft Power BI

Report

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1. AIM

To integrate advanced analytics with Business Intelligence (BI) reports and generate meaningful business insights, recommendations, and executive-level presentations using Microsoft Power BI.

2. OBJECTIVE

The objective of this experiment is to analyze the existing Power BI dashboards developed in Experiments 5 and 6, evaluate the overall business performance, identify strengths and weaknesses, explore business opportunities, and provide suitable recommendations for improving sales and profitability.

3. TOOLS AND TECHNOLOGIES USED

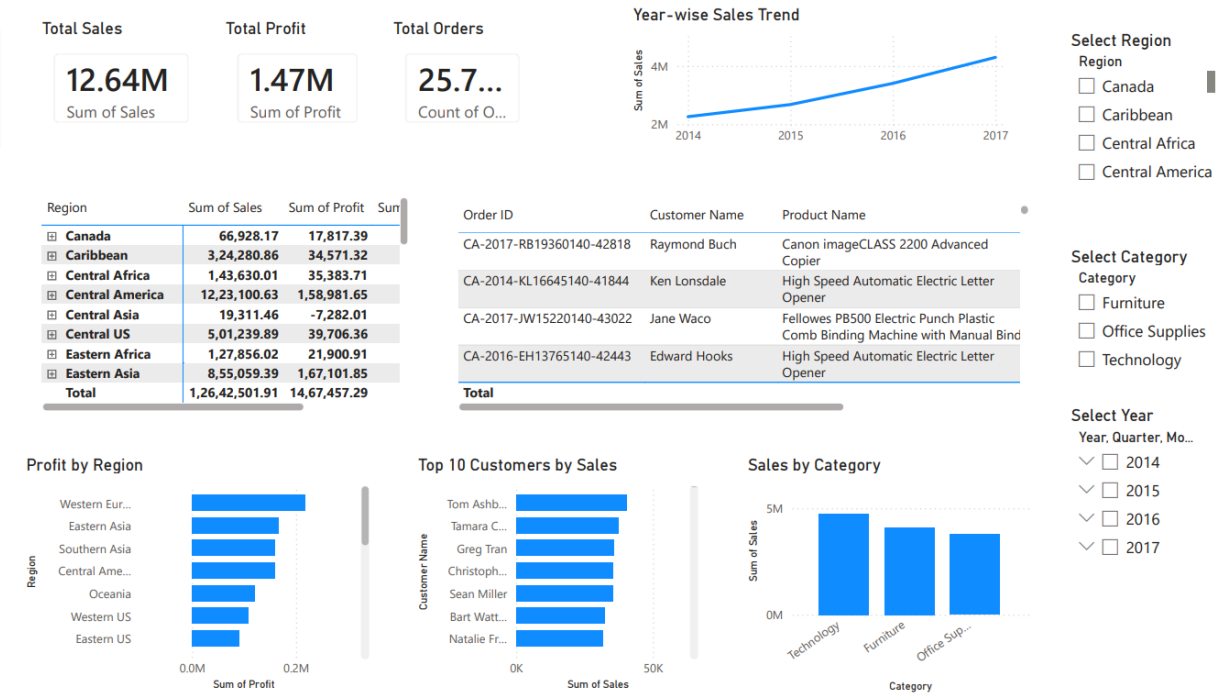
- Microsoft Power BI Desktop
- Global Superstore Dataset
- Power BI dashboards from Experiment 5 and Experiment 6
- Microsoft Word for report preparation
- Microsoft PowerPoint for executive presentation

4. REPORTS AND DASHBOARDS USED

The following dashboards were used for business analysis:

Experiment 5: Sales Analysis Dashboard

- Relational table containing Order ID, Customer Name, Product Name, Region, Sales, and Profit.
- Matrix visualization showing Region, State, and City hierarchy.
- Sales by Category.
- Profit by Region.
- Top 10 Customers by Sales.
- Year-wise Sales Trend.
- KPI cards showing Total Sales, Total Profit, and Total Orders.



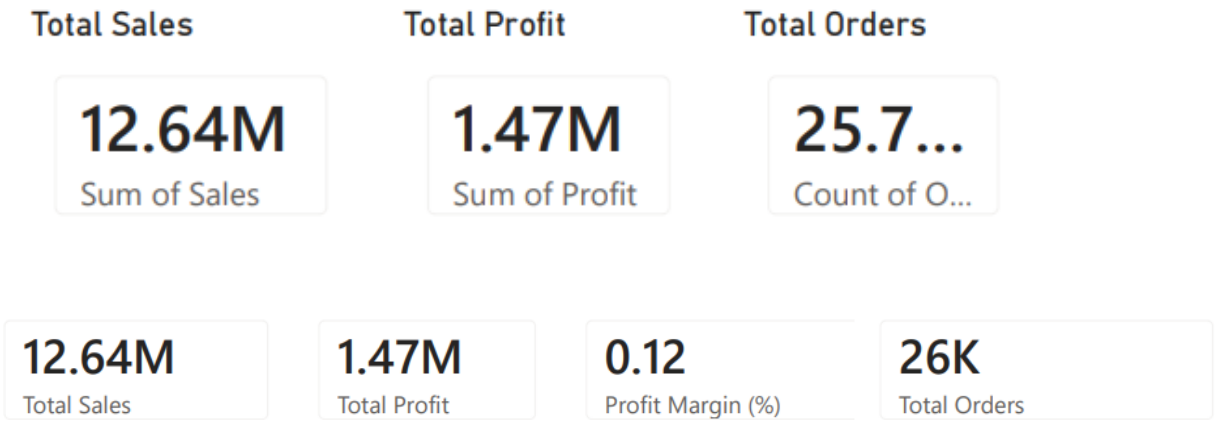
Experiment 6: Global Superstore Performance Dashboard

- Total Sales KPI.
- Total Profit KPI.
- Total Orders KPI.
- Profit Margin KPI.
- Sales by Region.
- Sales by Category.
- Monthly Sales Trend.
- Top 10 Customers by Sales.
- Profit by Segment.

- Interactive slicers for Region, Category, and Year.
- Scorecard comparing actual performance with targets.

5. KEY BUSINESS FINDINGS

The analysis of the existing dashboards provided the following findings:



5.1 Overall Business Performance

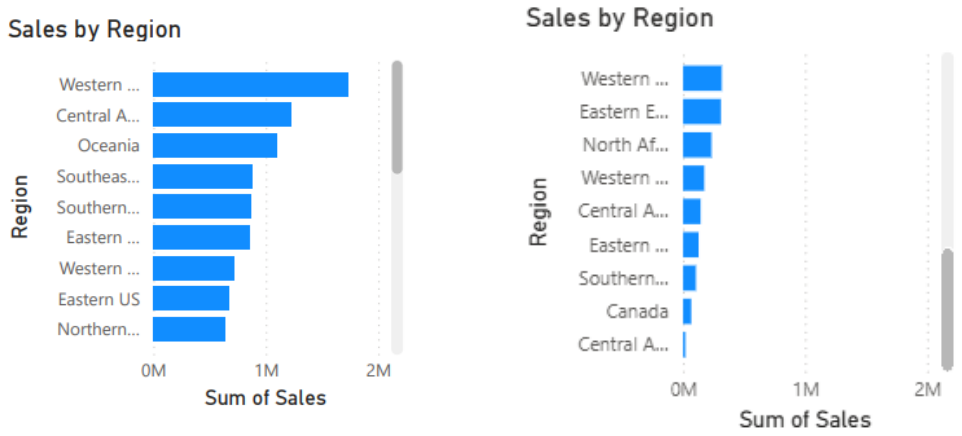
The business dashboard displays the following performance indicators:

KPI Value	Observed Value
Total Sales	12.64 Million
Total Profit	1.47 Million
Total Orders	26K
Total Margin	Approximately 12%

These indicators show that the business generates significant sales and profit. However, there is still scope to improve profitability.

5.2 Regional Performance

- Best Performing Region: Western Europe
- Worst Performing Region: Central Asia

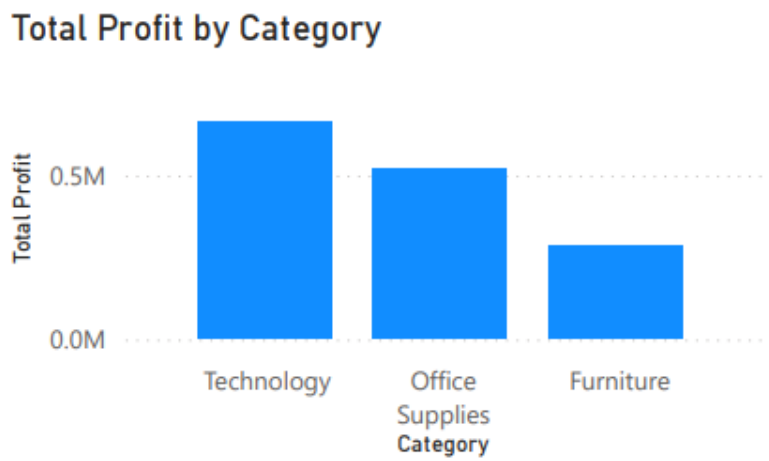


Western Europe contributes strongly to overall business performance, whereas Central Asia requires additional attention and improvement strategies.

5.3 Category-wise Profitability

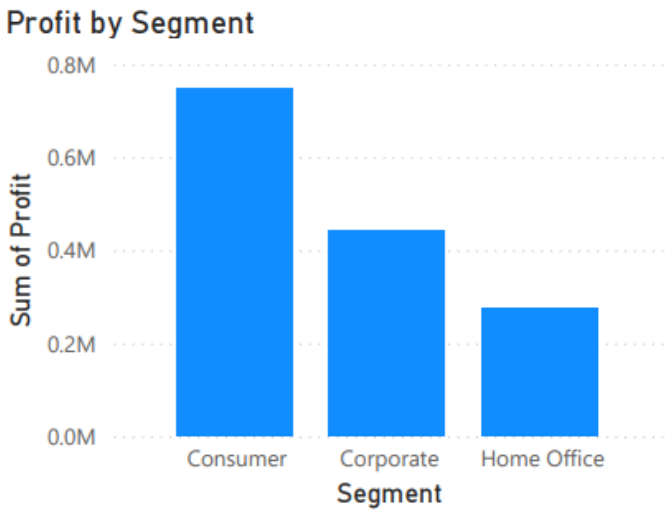
- Most Profitable Category: Technology
- Least Profitable Category: Furniture

Technology is the strongest category in terms of profitability. Furniture requires further analysis to identify factors affecting its profit performance.



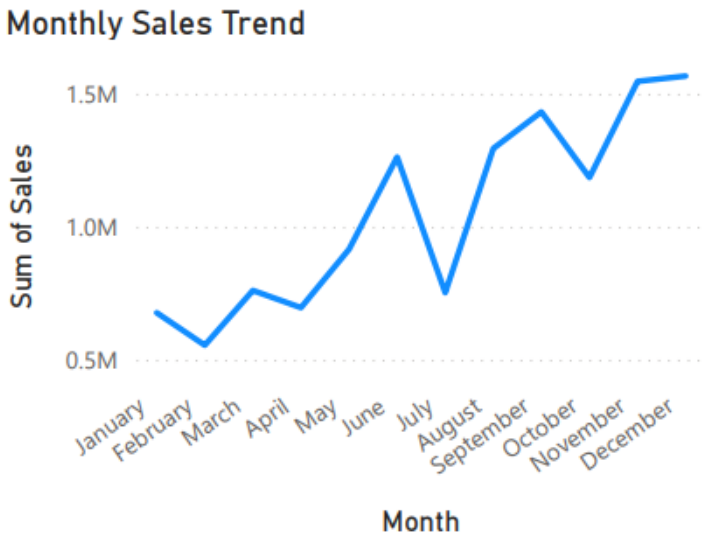
5.4 Customer Segment Performance

The dashboards provide a comparison of Consumer, Corporate, and Home Office customer segments based on sales and profit. This helps the business understand customer contribution and identify segments with growth potential.



5.5 Sales Trend Analysis

Monthly and yearly sales visualizations help identify changes in business performance over time. These trends can be used to understand seasonal variations and plan future sales strategies.



6. SWOT ANALYSIS

6.1 Strengths

1. Strong Overall Sales: The business has generated total sales of approximately 12.64 million.

2. Profitable Technology Category: Technology is the most profitable product category.
3. Strong Regional Performance: Western Europe is the best-performing region.
4. Data-Driven Monitoring: KPI cards and interactive dashboards allow management to monitor business performance.

6.2 Weaknesses

1. Low Profitability in Furniture: Furniture is the least profitable category and requires improvement.
2. Weak Regional Performance: Central Asia has the lowest regional performance and needs attention.
3. Profitability Improvement Required: Despite generating significant sales, the business has scope to improve its approximately 12% profit margin.

6.3 Opportunities

1. Expand Technology Sales: The business can increase investment in its most profitable category.
2. Improve Furniture Profitability: Pricing, discounting, and operational costs can be reviewed to improve margins.
3. Develop Low-Performing Regions: Targeted marketing and promotional strategies can help improve performance in Central Asia.
4. Customer Segment Growth: Customer segment analysis can help identify opportunities for targeted campaigns.

7. BUSINESS RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested:

1. **Focus on Profitable Categories:** Increase the promotion and availability of Technology products to improve overall profitability.
2. **Improve Furniture Performance:** Review pricing strategies, discounts, product costs, and inventory management.
3. **Develop Low-Performing Regions:** Introduce region-specific marketing campaigns and analyze customer demand in Central Asia.
4. **Improve Profit Margins:** Monitor discount levels and operating costs to ensure that higher sales also contribute to higher profits.
5. **Strengthen Customer Relationships:** Develop targeted offers and loyalty programs for valuable customer segments.

6. **Monitor Business Performance:** Regularly use Power BI dashboards and KPI scorecards to track sales, profit, and order performance.

9. CONCLUSION

In this experiment, the existing Power BI dashboards from Experiments 5 and 6 were analyzed to evaluate the overall performance of the Global Superstore business.

The analysis helped identify Western Europe as the best-performing region, Central Asia as the weakest-performing region, Technology as the most profitable category, and Furniture as the least profitable category.

The SWOT analysis highlighted the major strengths, weaknesses, and growth opportunities of the business. Based on these findings, recommendations were provided to improve category-wise profitability, regional performance, customer engagement, and overall profit margins.

Thus, the experiment demonstrated how Business Intelligence dashboards and advanced analytics can support effective business decision-making and strategic planning.

10. RESULT

Successfully analyzed existing BI dashboards, identified important business insights, performed SWOT analysis, and proposed recommendations for improving the overall business performance and profitability using Microsoft Power BI.